COMMERCIAL ITEM DESCRIPTION

ELECTROSURGICAL APPARATUS

The General Services Administration has authorized the use of this commercial item description.

This Commercial Item Description covers an electrosurgical apparatus in a reusable container, consisting of a solid state, radio frequency generator, operator controls and applicator devices.

Salient characteristics:

The unit shall be in accordance with UL 544 and ANSI/AAMI HF-18 Standards unless otherwise stated in this specification. All items shall conform to the applicable requirements of nationally recognized standards for the safety of medical equipment of the issue in effect on the date of solicitation. All items shall be approved as conforming to such a standard by an independent, nationally recognized testing laboratory.

The apparatus shall be a solid state electrosurgical unit (ESU) with monopolar and bipolar capability, suitable for use in surgical procedures.

Monopolar settings shall include a pure cut, pure coagulation, and a blend of cut and coagulation. Power output of the ESU shall remain relatively constant over the range of load impedance of 100 to 1000 ohms. In no instance shall maximum power exceed 400 watts unless clearly stated on a label attached to the ESU and in the operator documentation.

a. In a pure cut mode, the monopolar output power shall provide a minimum range of 0 to 300 watts, as measured into a manufacturer-specified resistive load that shall be between 100-500 ohms, inclusive. The unit shall be capable of delivering at least 150 watts of cutting power into 500 ohms.

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b. In the coagulation mode, the monopolar output power shall provide a minimum range of 0 to 100 watts, as measured into a manufacturer-specified resistive load that shall be between 100-500 ohms, inclusive. The unit shall be capable of delivering at least 50 watts of coagulation power into 500 ohms and exhibit minimal cutting characteristics when operating in this mode.

c. The blend mode shall provide cutting waveforms with a choice of at least three degrees of hemostasis, or allow variation of the relative amounts of hemostasis and cutting at any power setting by means of stepped or stepless controls

Bipolar output control shall be independent of monopolar controls, and shall produce a coagulation waveform. Bipolar output power shall be adjustable and provide a minimum range of 0 to 30 watts, as measured into a manufacturer-specified resistive load that must be between 50-300 ohms, inclusive. Maximum output power shall not exceed 100 watts, unless a label clearly states otherwise.

Open circuit voltage of the unit shall be high enough to allow the surgeon to fulgurate (spray coagulate) tissue. In the coag mode, open circuit voltage, peak-to-peak, shall be at least 5000. The unit shall withstand repeated and prolonged open circuit voltage.

The electrosurgical apparatus shall have separate, adjustable cut, coagulate, and bipolar output power level controls. The controls may be stepped or continuously variable. At the minimum power setting, there shall be zero output power at zero control setting. Power increase shall be incremented with a maximum of 5 watt increments.

The use of momentary activation switches for selection of power level is acceptable. If momentary activation switch is provided, the unit must be set to zero power level or "standby" mode at initial power up.

Output shall be activated by means of a footswitch, a hand control style electrode pencil, and forceps with a built-in switch.

a. Footswitch activation. Footswitch shall be splashproof, grounded, and have suitable mechanical protection against accidental depression. Shall be designed so that cut mode output is activated by the leftmost control of the footswitch as viewed by the operator.

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b. Handpiece actuation. The hand control style electrode pencil shall provide the capability for the user to select between the coagulation and cutting modes at the handpiece in the monopolar mode. If two switches are provided, the switch nearer the active electrode shall activate the cutting mode. The "cut" switch shall be color coded yellow and the "coagulation" switch shall be color coded blue. With rocker or sliding types of switch mechanisms, a forward motion (toward the electrode) or pressure on the forward part of the switch mechanism shall activate the cutting mode.

c. Forceps actuation. Monopolar or bipolar outputs shall be switch-selectable by means of a switch built into the forceps.

The cutting and coagulation switches shall not be able to be activated simultaneously.

Output power, as measured at the active electrode, shall remain constant within 10 percent over the full tolerance range of line voltages specified in the paragraph titled "Electrical requirements.".

The unit shall have isolated output circuits; i.e. output not referenced to ground.

The unit shall generate minimal radio frequency interference (RFI), both radiated and conducted, through the power line. The active cable of the ESU shall be properly insulated to limit radiated losses and prevent capacitive coupling into nearby wires and metal equipment.

The unit shall be provided with a positive or self-latching on/off switch. An indicator light shall signal that the ESU has been turned on. Separate visual indicators shall be incorporated to indicate when the ESU "RF" output has been activated. In addition, a minimum of two distinct audible tones shall indicate cut, and coag/bipolar output activation. An adjustable volume control which cannot reduce the sound level below 45 dB shall be provided.

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Front panel controls shall be visible and easily identifiable from a distance of at least 8 feet for vision corrected to 20/20. Footswitch and electrode connections shall be clearly labeled, color coded, or size keyed to prevent accidental attachment to improper outlets. Control indicators shall be color coded. Controls and output connectors shall be arranged in an orderly sequence. Condensed operating instructions and warning placards shall be mounted on the unit.

To prevent patient burns, the dispersive electrodes (disposable and reusable) shall disperse the return current over an adequately large area. A reusable dispersive electrode shall be supplied which shall conform to the patient, resist scratching and permanent bending, and shall be easy to clean. The electrodes shall not be adversely affected by surgical prep solutions or physiological fluids.

The unit shall have a dispersive electrode monitoring system to monitor the contact quality between the patient and the dispersive electrode.

A detection system shall be provided to monitor the electrical continuity of the current path from dispersive electrode to the ESU. If a break occurs in this pathway, the ESU shall de-energize and activate a fault indicator until the unit's return circuit is corrected.

The unit shall be convection cooled. A fan, which shall be thermally activated, may be incorporated to provide adequate component cooling and to dissipate internal heat. If a fan is provided, placement of ventilation openings shall deflect exhaust air downward and away from the sterile field. Position of vents shall not allow accidental contact with electrical connections or components, and shall be designed to restrict entry of spilled fluids.

Overcurrent protection for the ESU, preferably in the form of a circuit breaker, is required. If a fuse is used, a spare shall be provided and clearly labeled as to function and rating. Circuit breakers shall be protected from other objects impinging against them.

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The unit shall be of rugged construction capable of withstanding hazards of normal use in a hospital setting. It shall pass all the National Safety Transit Association preshipment test procedures.

Electrical requirements:

a. The electrosurgical apparatus shall be capable of operating in all of the four conditions listed below. There shall be an easily accessible switch which shall control the conditions listed below.

1. 115 VAC plus or minus 10 percent, 50 plus or minus 2 Hz.
2. 115 VAC plus or minus 10 percent, 60 plus or minus 2 Hz
3. 230 VAC plus or minus 10 percent, 50 plus or minus 2 Hz
4. 230 VAC plus or minus 10 percent, 60 plus or minus 2 Hz.

b. Change over devices, switches, circuits, or transformers if used for the purpose of voltage or frequency conversion, shall be clearly marked and identified in the appropriate documentation which accompanies the unit. Mechanical or electrical adjustments to effect the conversion shall not require the use of specialized tools or skills.

Minimum length of the cords and cables supplied shall be:

Cable Type	Length
Line Cord	9.8 feet
Active	8.2 feet
Dispersive	7.9 feet
Footswitch	9.8 feet

Each unit shall be supplied with all accessories necessary for a fully functional system. As a minimum, the following accessories shall be included:

a. One mobile cart. The cart shall provide storage space for the footswitch and other accessories. When the ESU is mounted on the cart, the front panel controls shall be highly visible and easily identifiable.

b. Twelve disposable dispersive electrodes with cord.

c. **Twelve disposable dispersive electrodes, split plate, full surface conductor, with preattached cord.**

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d. One disposable hand activated pencil, which shall be presterilized and include a one inch blade, cord and connector.

e. One reusable dispersive electrode.

f. One dispersive electrode adapter, if necessary, having a one inch blade, cord and connector

g One reusable hand activated handle, presterilized, which accepts standard 3/32 inch shaft electrodes, having a removable one inch blade, cord, and connector.

h One reusable foot activated handle, which accepts standard 3/32 inch shaft electrodes, with cord.

i. One bipolar forceps, hand switching, 6-1/2 inch or 7 inch, straight tip, with cord.

j One forceps, Hudson (Ewald), switching bipolar, 4-3/4 inch straight tip.

k. One (or two) footswitch(es), for monopolar and bipolar activation.

l. Twelve tubes electrosurgical gel, 8 ounces per tube.

m. Twelve electrodes, 1 inch, straight blade.

n. One electrode, 7/32 inch, ball.

o One electrode, 3/4 inch or 1 inch, needle.

p One electrode 3/32 inch, angled ball.

q. One electrode, 3/8 inch, round loop.

Operation and Service Data Manuals for Medical and Laboratory Instrumentation.

a Information content:

The contractor shall furnish manuals, handbooks and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures or illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures

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shall be written in the English language and all schematics and wiring diagrams shall use American electrical and electronics symbols. The manuals shall include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name, and quantity required. All service notes, service memos, etc. pertaining to the equipment and issued subsequent to the printing of the original instructions shall be, as a minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment the service data shall include complete instructions and drawings which show interfacing of all system components. The operator's/instruction manual shall include a listing of cautionary warnings of the potential hazards of electrosurgical devices, including: electric shock, thermal burns, fire and explosion, cataract formation, pacemaker interference, RF - interference and irradiation, cardiac arrhythmias, and allergic or toxic reaction to materials. The operator's manual shall also state the recommended method of sterilization of all required sterile reusable components.

b. Quantities, distribution and schedules:

Two complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of two service manuals need to be supplied to each consignee. However, two operation manuals still need to be supplied with each unit.

Workmanship. The electrosurgical apparatus shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (Ea). One electrosurgical apparatus, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of S-2. The unit of product for examination purposes shall be one electrosurgical apparatus.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of S-2. The unit of product for test purposes shall be one electrosurgical apparatus.

Tests Tests shall be conducted to determine compliance with specification requirements. The entire unit shall be assembled with all components properly connected. Testing shall include but shall not be limited to the following:

Input power test. Verify the unit operates on each specified voltage and Hz. All remaining tests are to be performed using each specified input power.

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Output power test. Verify the following:

a. The monopolar output power, at any setting, peaks at a load impedance between 100 and 1000 ohms.

b. In a pure cut mode, the monopolar output power provides a minimum range of 0 to 300 watts as measured into a resistance load specified by the manufacturer (must be between 100 - 500 ohms).

c. The unit is capable of delivering at least 150 watts of cutting power into 500 ohms.

d. In the coagulation mode, the monopolar output power provides a minimum range of 0 to 100 watts as measured into a resistance load specified by the manufacturer (must be between 100 - 500 ohms).

e. The unit provides coagulation power of a least 50 watts and exhibits minimal cutting characteristics when operating in this mode.

f. In the bipolar mode, output power provides a minimum range of 0 to 30 watts as measured into a resistance load specified by the manufacturer (must be between 50 - 300 ohms).

Maximum output power test. Verify the maximum power in monopolar settings into any load resistance does not exceed 400 watts, unless a label clearly states otherwise. Verify that maximum output in bipolar settings does not exceed 100 watts, unless a label clearly states otherwise.

Power control sensitivity test. Verify the power increase does not exceed 5 watt increments for each step of a stepped control.

Oper circuit voltage test. Verify that in the coag mode, the open-circuit voltage, peak to peak, is at least 5000 volts into a 500 ohm load.

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Minimum power control setting test.

a. For units with stepped or continuously variable power level controls, verify at the minimum power setting there is zero output power at the zero control setting.

b For units with momentary activation switches, verify the unit is set to zero power level or "standby" mode at initial power-up.

Line voltage fluctuation test. Verify that the output power measured at the active electrode remains constant within 10 percent when the line voltage is fluctuated over the tolerance range specified for each voltage/frequency combination specified under salient characteristics paragraph titled "Electrical requirements".

Visual indicators test. Verify the unit provides separate visual indicators for input power turned on and for output activated.

Audible indicators test. Verify the unit provides distinct audible tones to indicate cut, and coag/bipolar activation. Verify the volume cannot be reduced below 45 dB.

Output isolation test Verify the power delivered to a 200 ohm load connected between each active output and ground does not exceed 4.5 watts at maximum output power setting.

Radio frequency leakage test. Verify the power delivered to a 200 ohm non-reactive resistance connected from the dispersive electrode to ground does not exceed 4.5 W at maximum output power setting with the electrodes operated open-circuited.

Electrical safety test. Verify no more than 500 uA of sink current flows to ground with 110 V, 60 Hz applied to the active or return electrode.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

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If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

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Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

6515-01-314-1237

MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA-MPQAS
USPHS
VA - OSS

Project No. 6515-5269

Location: ESUMC16.DOC

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-54779	2. DOCUMENT DATE (YYMMDD): 15 January 1993

3. DOCUMENT TITLE: ELECTROSURGICAL APPARATUS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial: – (2) AUTOVON: (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME: DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870 (2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340, AUTOVON 289-2340

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